# Manipulating Surds

Exam code: 9MA0

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Written by: Roger B
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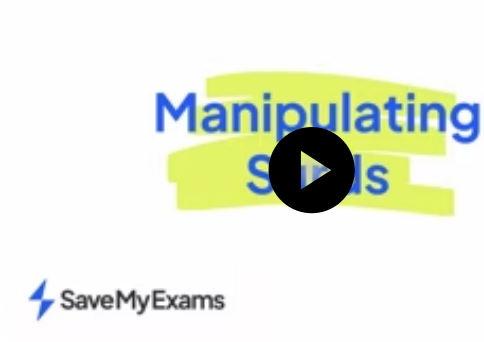


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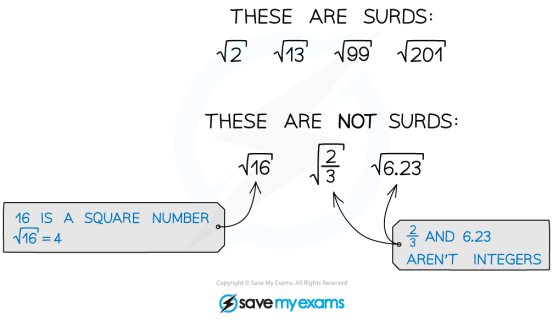
## Manipulating surds

### What are surds?

- If $n$ is a positive integer and is not a square number, then $\sqrt{n}$ (or any multiple of $\sqrt{n}$ ) is a **surd**




- Surds are examples of irrational numbers

### What do I need to know about manipulating surds?

- There are two basic rules you need to know in order to manipulate and simplify surds:

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$

DON'T FORGET THAT THIS SECOND LAW IS THE SAME THING AS

$$\sqrt{a \div b} = \sqrt{a} \div \sqrt{b}$$

ALSO DON'T FORGET HOW THIS WORKS WITH RECIPROCALS:

$$\frac{1}{\sqrt{\frac{a}{b}}} = \frac{1}{\left(\frac{\sqrt{a}}{\sqrt{b}}\right)} = \frac{\sqrt{b}}{\sqrt{a}} = \sqrt{\frac{b}{a}}$$

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- When simplifying, look for square factors

LOOK FOR FACTORS THAT ARE SQUARE NUMBERS

THE ANSWERS HERE $\sqrt{8}$ AND $\sqrt{720}$ IN SIMPLEST SURD FORM

$$\sqrt{8} = \sqrt{4 \times 2} = \sqrt{4} \times \sqrt{2} = 2\sqrt{2}$$

$$\sqrt{720} = \sqrt{144 \times 5} = \sqrt{144} \times \sqrt{5} = 12\sqrt{5}$$

- You can collect like terms with surds like you do with letters in algebra:

$$7\sqrt{3} - 5\sqrt{3} = 2\sqrt{3}$$

$$5\sqrt{2} - 2\sqrt{5} - 8\sqrt{2} + 3\sqrt{5} = -3\sqrt{2} + \sqrt{5}$$

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- But be careful not to confuse this with the multiplication and division rules... cannot add or subtract 'under the surd':

$\sqrt{a} + \sqrt{b}$ IS NOT EQUAL TO $\sqrt{a+b}$

$\sqrt{a} - \sqrt{b}$ IS NOT EQUAL TO $\sqrt{a-b}$

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## Where can surds be useful?

- Using surds lets you leave answers in exact form
  - e.g. $5\sqrt{2}$ rather than $7.071067812$
- Understanding how to simplify surds can help reduce expressions and collect like terms

you need to carry an exact value through to another part of working

- When simplifying surds, remembering your square numbers is really helpful!

## Worked Example

Simplify $\sqrt{125} - \sqrt{45}$ giving your answer in the form $a\sqrt{5}$, where $a$ is an integer.

THE FORM OF THE REQUIRED ANSWER LETS YOU KNOW THAT 5 SHOULD BE ONE OF THE FACTORS IN BOTH SURDS HERE!

$$\sqrt{125} - \sqrt{45} = \sqrt{25 \times 5} - \sqrt{9 \times 5}$$
$$= \sqrt{25} \times \sqrt{5} - \sqrt{9} \times \sqrt{5}$$
$$= 5\sqrt{5} - 3\sqrt{5}$$
$$= 2\sqrt{5}$$

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